**This assignment will not be graded and is only for practice.**

## 2.1 Level 1:

1) Suppose $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 11 & -2 & 9 & -6 \\ -3 & 4 & 7 & 7 \end{bmatrix}$. Which of the following is true about the matrix $A$? ***1 point***

[Hint: The $(i, j)$-th entry is the entry which is at the $i$-th row and $j$-th column.]

- [ ] It is a $4 \times 3$ matrix.
- [x] It is a $3 \times 4$ matrix.
- [ ] $(2, 3)$-th entry of the matrix $A$ is $4$.
- [x] $(2, 3)$-th entry of the matrix $A$ is $9$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

It is a $3 \times 4$ matrix.
$(2, 3)$-th entry of the matrix $A$ is $9$.

2) Which of the following statements is(are) TRUE? ***1 point***

[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

- ( ) Any diagonal matrix is a scalar matrix.
- ( ) Scalar matrices may not be square matrices.
- (•) Scalar matrices must be square matrices.
- ( ) Any scalar matrix is an identity matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Scalar matrices must be square matrices.

3) ***1 point***

Given below is the system of linear equations:

$$7x_1 + 10x_2 + 12x_3 = 36$$
$$8x_1 + 4x_2 - 9x_3 = 11$$
$$4x_1 - x_2 + 3x_3 = 10$$

If the matrix representation of the system of equations is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

- (●) $A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- ( ) $A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & 9 & 11 \\ 4 & 1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- ( ) $A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & -9 & 11 \\ 4 & -1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- ( ) $A = \begin{bmatrix} 7 & 10 & 36 & 12 \\ 8 & 4 & 11 & -9 \\ 4 & -1 & 10 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$

4) Which of the following statements is(are) TRUE? ***1 point***

- [ ] Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.
- [x] Addition of two matrices is possible only if the orders of both the matrices are the same.
- [ ] Defining $AB$ is possible if the number of rows in the matrix $A$ is same as the number of columns in the matrix $B$.

- [x] Defining $AB$ is possible if the number of columns in the matrix $A$ is same as the number of rows in the matrix $B$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Addition of two matrices is possible only if the orders of both the matrices are the same. Defining $AB$ is possible if the number of columns in the matrix $A$ is same as the number of rows in the matrix $B$.

5) Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}, Q = \begin{bmatrix} 1 & 4 & -9 \end{bmatrix}, R = \begin{bmatrix} 0 & -3 & 10 \end{bmatrix}, D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$ ***1 point***

[Hint: If $A$ is a matrix of order $m \times n$ and $B$ is a matrix of order $n \times p$, then the order of $AB$ is $m \times p$.]

- [x] The matrix $PD$ is of order $3 \times 1$.
- [ ] The matrix $PD$ is of order $1 \times 3$.
- [ ] The matrix $QD$ is of order $3 \times 3$.
- [x] The matrix $QD$ is of order $1 \times 1$.
- [x] The matrix $DQ$ is of order $3 \times 3$.
- [ ] The matrix $DQ$ is of order $1 \times 1$.
- [ ] The product $QD$ is not defined.
- [x] The product $QR$ is not defined.
- [x] The addition $P + Q$ is not defined.
- [x] The addition $P + D$ is not defined.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix $PD$ is of order $3 \times 1$.

The matrix $QD$ is of order $1 \times 1$.
The matrix $DQ$ is of order $3 \times 3$.
The product $QR$ is not defined.
The addition $P + Q$ is not defined.
The addition $P + D$ is not defined.

**2.2 Level 2:**

6) Choose the set of correct options. ***1 point***

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given conditions to find matrix $A$, for first two options.]

- [ ] If $A$ is a square matrix of order 2 and $A^2 = I$, then $A = I$ or $A = -I$ where $I$ is the identity matrix of order 2.
- [ ] If $A$ is a square matrix of order 2 and $A^2 = 0$, then $A = 0$.
- [x] If $A$ and $B$ are square matrices of order 3 and $A + B = 0$, then $B = -A$.
- [x] If $A$ is a scalar matrix of order 3, $B$ is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If $A$ and $B$ are square matrices of order 3 and $A + B = 0$, then $B = -A$.
If $A$ is a scalar matrix of order 3, $B$ is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

7) If $A$ is a square matrix of order $2$ whose first column is denoted by $C_1$ and second column is denoted by $C_2$ (the left most column is taken as the first one) and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ then choose the set of correct options. ***1 point***

[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]

- [x] The first column of $AB$ is $b_{11}C_1 + b_{21}C_2$.
- [x] The second column of $AB$ is $b_{12}C_1 + b_{22}C_2$.
- [ ] The first column of $AB$ is $b_{21}C_1 + b_{11}C_2$.
- [ ] The second column of $AB$ is $b_{22}C_1 + b_{21}C_2$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The first column of $AB$ is $b_{11}C_1 + b_{21}C_2$.
The second column of $AB$ is $b_{12}C_1 + b_{22}C_2$.

8) Choose the set of correct options. ***1 point***

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given condition to find matrix $A$ for last two options.]

- [x] There exist some real matrices $A$ and $B$, such that $AB = BA$.
- [ ] There do not exist any real matrices $A$ and $B$, such that $AB = BA$.
- [ ] There does not exist any real matrix $A$, such that $A^2 = A$.
- [x] There exists some real $2 \times 2$ matrix $A$, such that $A^2 + A + I = 0$

Yes, the answer is correct.

Score: 1

Accepted Answers:

There exist some real matrices $A$ and $B$, such that $AB = BA$.
There exists some real $2 \times 2$ matrix $A$, such that $A^2 + A + I = 0$

9) Suppose $A$ is a $3 \times 3$ scalar matrix and (1,1)-th entry of the matrix $A$ is 4. Suppose $B$ is a $3 \times 3$ square matrix such that $(i, j)$-th entry is equal to $i^2 + j^2$. Find the (2,2)-th entry of the matrix $2A + B$.

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Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 16

***1 point***

10) Suppose $A = \begin{bmatrix} -1 & 2 \\ -4 & 7 \end{bmatrix}$ and $A^2 - \alpha A + I = 0$ for some $\alpha \in \mathrm{R}$.

Find the value of $\alpha$.

6

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 6

***1 point***

Your score is: 10/10